

Cover Shelter Design

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1 Input Variables

The following sets and parameters are defined as input data to the model:

- V : Set of vertices (container locations).
- T : Set of container types.
- $r \in T$: Residential container type.
- C_t : Maximum number of residential vertices that can be assigned to a building of type $t \in T \setminus \{r\}$.
- D_t : Service distance threshold for type $t \in T \setminus \{r\}$, indicating the maximum distance within which the type can serve residential vertices. Distances follow a mixed Manhattan–Euclidean metric.
- S_t : Number of containers that constitute one complete group of type $t \in T \setminus \{r\}$.
- Δ_t : Maximum allowable distance between any two vertices belonging to the same group of type $t \in T \setminus \{r\}$.
- $d(u, v)$: Mixed Manhattan–Euclidean distance between vertices $u, v \in V$.
- $h(u, v)$: Euclidean distance between vertices $u, v \in V$, used for compactness or clustering penalties.

2 Precomputed Variables

The following auxiliary sets are precomputed based on spatial relationships between vertices and type-specific parameters.

Candidate Subgroups:

$$U(t) = \left\{ U \subseteq V \mid |U| = S_t \right\}, \quad \forall t \in T \setminus \{r\}.$$

Explanation: For each non-residential type t , the set $U(t)$ contains all possible subsets of vertices of size S_t . Each subset represents a potential configuration (subgraph) of containers that could form one complete group of type t .

Feasible Groups Under Diameter Constraint:

$$G_t = \left\{ U \in U(t) \mid d(u, v) \leq \Delta_t, \forall u, v \in U \right\}, \quad \forall t \in T \setminus \{r\}.$$

Explanation: The subset G_t filters $U(t)$ by removing all candidate groups whose internal pairwise distances exceed the maximum allowable limit Δ_t . Thus, G_t includes only compact, spatially feasible groups for each non-residential type.

Neighborhood Groups for Residential Vertices:

$$S_{v,t} = \left\{ G \in G_t \mid \exists u \in G : d(u, v) \leq D_t \right\}, \quad \forall v \in V, \forall t \in T \setminus \{r\}.$$

Explanation: For each vertex v and type t , the set $S_{v,t}$ contains all feasible groups of type t that are within the service distance D_t from v . Intuitively, it defines which type- t facilities (groups) can potentially serve residential container v under the mixed Manhattan–Euclidean distance metric.

3 Decision Variables

All decision variables are binary.

Type Assignment Variables:

$$x_{v,t} = \begin{cases} 1, & \text{if vertex } v \\ & \text{is assigned type } t, \\ 0, & \text{otherwise.} \end{cases} \quad \forall v \in V, \forall t \in T.$$

Explanation: The variable $x_{v,t}$ indicates whether vertex v hosts a container of type t . Each vertex can take at most one type, including the residential type r .

Residential-to-Group Assignment Variables:

$$y_{v,g,t} = \begin{cases} 1, & \text{if residential vertex } v \\ & \text{is assigned to group } g \\ & \text{of type } t, \\ 0, & \text{otherwise.} \end{cases} \quad \forall v \in V, \forall g \in G_t, \forall t \in T \setminus \{r\}.$$

Explanation: The variable $y_{v,g,t}$ links each residential vertex to a feasible group g of a non-residential type t . It determines whether vertex v is served by that particular facility group.

Group Activation Variables:

$$u_{g,t} = \begin{cases} 1, & \text{if group } g \\ & \text{of type } t \text{ is activated,} \\ 0, & \text{otherwise.} \end{cases} \quad \forall g \in G_t, \forall t \in T \setminus \{r\}.$$

Explanation: The binary variable $u_{g,t}$ activates or deactivates group g of type t . Only activated groups can serve residential vertices and contribute to the model's coverage and capacity constraints.

4 Objective

Maximize the total number of residential containers (where r denotes the residential type):

$$\max \sum_{v \in V} x_{v,r}$$

5 Core Constraints

1. Single type per vertex:

$$\sum_{t \in T} x_{v,t} \leq 1 \quad \forall v \in V.$$

Explanation: Each vertex can host at most one container type, including the residential type r .

2. Neighborhood coverage (Rainbow constraint):

$$\sum_{g \in S_{v,t}} u_{g,t} \geq x_{v,r} \quad \forall v \in V, \forall t \in T \setminus \{r\}.$$

Explanation: For every residential vertex v , and for each non-residential type t , at least one activated group $u_{g,t}$ within $S_{v,t}$ must exist within service distance D_t . This ensures that every residential vertex is covered by all required non-residential facilities.

3. Group size consistency:

$$u_{g,t} \leq x_{v,t} \quad \forall t \in T \setminus \{r\}, \forall g \in G_t, \forall v \in g.$$

Explanation: A group g of type t can only be activated if all its member vertices v are assigned as that same type. This enforces internal consistency between group activation and vertex-type assignment.

4. Group-vertex membership coherence:

$$x_{v,t} \leq \sum_{\substack{g \in G_t \\ v \in g}} u_{g,t}, \quad \sum_{\substack{g \in G_t \\ v \in g}} u_{g,t} \leq 1, \quad \forall v \in V, \forall t \in T \setminus \{r\}.$$

Explanation: A vertex v can be assigned to type t only if it belongs to at least one active group of that type, and each vertex can participate in at most one such group. These two inequalities jointly prevent cross-group overlaps and maintain group exclusivity.

5. **Assignment equality constraint:**

$$\sum_{g \in S_{v,t}} y_{v,g,t} = x_{v,r}, \quad \forall v \in V, \forall t \in T \setminus \{r\}.$$

Explanation: This equality enforces that every residential vertex v must be assigned to exactly one active group g of each non-residential type t . If $x_{v,r} = 1$, the vertex must appear in one and only one assignment $y_{v,g,t} = 1$. If $x_{v,r} = 0$, no assignment can exist for that vertex. This ensures that each residential container receives full coverage and activates the relevant demand and capacity constrain

6. **Demand (capacity) constraint:**

$$\sum_{v \in V} y_{v,g,t} \leq C_t u_{g,t} \quad \forall t \in T \setminus \{r\}, \forall g \in G_t.$$

Explanation: This constraint enforces the demand–capacity relationship between residential vertices and non-residential groups. For each group g of type t , the left-hand side represents the total number of residential vertices assigned to that group. The right-hand side, $C_t u_{g,t}$, limits this number by the capacity parameter C_t . If the group is inactive ($u_{g,t} = 0$), no residential vertex can be assigned to it. If the group is active ($u_{g,t} = 1$), it may serve up to C_t residential vertices but not more. This ensures that the total service load of each non-residential group never exceeds its predefined maximum capacity.

Linking constraints between assignments and activations:

$$y_{v,g,t} \leq u_{g,t}, \quad y_{v,g,t} \leq x_{v,r}, \quad \forall v \in V, \forall t \in T \setminus \{r\}, \forall g \in S_{v,t}.$$

Explanation: These linking constraints preserve logical consistency between different decision variables:

- The inequality $y_{v,g,t} \leq u_{g,t}$ ensures that a residential vertex v can only be assigned to a group g if that group is activated.
- The inequality $y_{v,g,t} \leq x_{v,r}$ guarantees that only vertices designated as residential ($x_{v,r} = 1$) can receive service from a non-residential group.

Together, these two relationships prevent infeasible assignments such as linking inactive groups to residential vertices or assigning non-residential vertices to any facility group.